

DAMA Chapter 4: Data Architecture

Presentation overview with concepts, examples, and visual summaries

Prepared for Data Management Learning

Based on *DAMA-DMBOK2*, Chapter 4

This deck is based on the uploaded *DAMA-DMBOK: Data Management Body of Knowledge, 2nd Edition*, especially Chapter 4 on Data Architecture.

- Understand what DAMA means by **data architecture** and why it matters.
- See how data architecture links **business strategy**, **data requirements**, and **technology execution**.
- Recognize core artifacts: **enterprise data models**, **subject area models**, and **data flows**.
- Identify practical activities, governance needs, and success factors for implementation.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

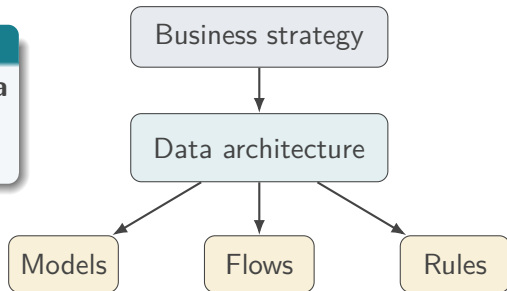
What is data architecture?

Core idea

Data architecture is the **blueprint for managing data assets**. It describes how data is organized, defined, moved, controlled, and aligned with enterprise goals.

- Covers artifacts, activities, and behaviors
- Works across multiple levels of abstraction
- Helps people understand data that is too large and complex to grasp directly

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).



Why organizations need it

- Bridges **business strategy** and **technology execution**
- Helps organizations evolve products, services, and data faster
- Makes sure processes consistently get the data they need
- Improves alignment between **business** and **IT**
- Supports agility, integration, and controlled change

Realistic examples

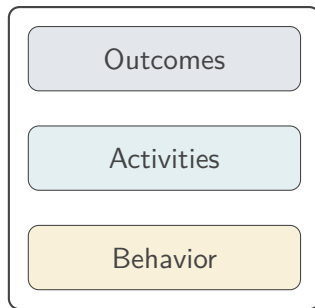
- Bank launching a new digital lending journey
- Hospital integrating patient, lab, and billing data
- Manufacturer shifting from selling products to *pay-per-use* services

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

DAMA's 3-part view of data architecture

Three components

- 1 **Outcomes** – models, definitions, data flows, roadmaps
- 2 **Activities** – forming, deploying, and maintaining architecture
- 3 **Behavior** – collaboration, mindset, and architectural discipline



Together form data architecture

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Key outcomes and practice goals

- Define the **current state** of organizational data
- Provide a **standard business vocabulary** for data and components
- Align data architecture with **enterprise strategy** and **business architecture**
- Express **strategic data requirements** and high-level integrated designs
- Connect architecture work to the overall enterprise roadmap

What good practice looks like

Use architecture artifacts as master blueprints, influence projects and stakeholders, and establish the semantics of the enterprise through shared vocabulary.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Architecture works across domains

Domain	Main focus	Example question
Business architecture	How the enterprise creates value	Which capabilities and processes matter most?
Data architecture	How data should be organized and managed	What data do we need, and how should it fit together?
Applications architecture	Structure and function of applications	Which systems create or consume the data?
Technology architecture	Platforms, networks, and infrastructure	What technical environment hosts and protects the systems?

Takeaway

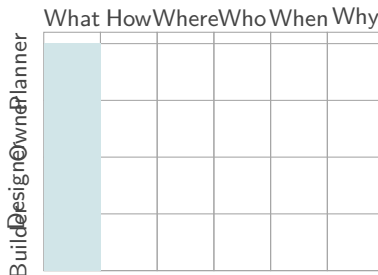
Data architecture does not stand alone. It must work with business, application, and technology architecture.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Enterprise architecture frameworks

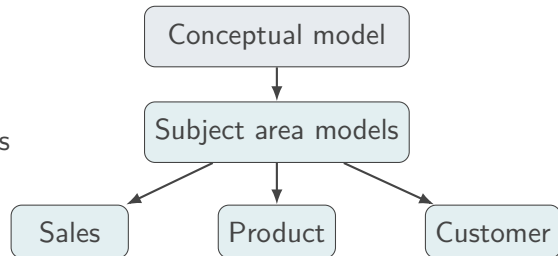
- Frameworks help non-architects understand architecture relationships
- DAMA references common enterprise architecture frameworks
- The most visible example in the chapter is the **Zachman Framework**
- Zachman shows what models should exist from multiple stakeholder perspectives

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).



Enterprise data model: the master blueprint

- The most detailed architecture design document is the **enterprise data model**
- It contains names, definitions, conceptual and logical entities, relationships, and business rules
- It is built across levels: conceptual → subject area → logical detail
- Physical models belong more directly to data modeling and design



Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Subject area models make the enterprise understandable

- Each entity should live in one subject area, but can relate to entities in other areas
- Subject areas can be formed using normalization rules, ownership structures, value chains, or business capabilities
- DAMA notes that a **combined top-down and bottom-up approach** is often practical

Example subject areas

Customer, Product, Sales Order, Inventory, Supplier, Billing, Service, Claims.

Teaching example

In retail, *Customer* belongs in one core subject area, but connects to orders, returns, support cases, and loyalty events across the enterprise.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Data Architecture Implementation

- Implementation refers to putting Data Architecture into action
- Implementation of Data Architecture includes the below key points:
 - Creation of Data models, data flows and roadmaps
 - Setting up of teams
 - Setting up the way of working
 - Using data architecture in project
 - Creating awareness in the organization
- Implementation usually starts in the project and grows across the organization.
- Early project may need extra efforts and funding

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Key elements in Implementation

Below are the central elements that are involved in implementation of Data Architecture:

① Metadata

- Data Architecture artifacts such as models, flows are crucial metadata
- Metadata helps understand data and system
- It is used to validate the schema

② Governance Basics

- It is helpful in defining rules, standards and compliance
- It helps ensure project flow architecture

③ Data Quality

- It aims at improving the data quality and trust
- It ensures that inconsistent data and data copies are avoided

① Master Data (MDM Concept)

- Implementation of a Data Architecture mostly starts with master data(e.g., customer, product)
- This is then expanded to transaction data

② Architectural Patterns

There are mainly two approaches:

- Agile: step by step, incremental and flexible
- Traditional: Blueprints, standards are followed

However, mostly in the organization the hybrid approach is followed which involves both mentioned above

③ Data Architecture ensures the below:

- Standard structure(Standardization)
- Integration across system (Integration)
- Reuse of data assets (Re-usability)

① Risk: The main risk involved in building a Data Architecture are:

- Lack of Management support
- Poor communication
- Resistance to change
- Inexperienced project leaders

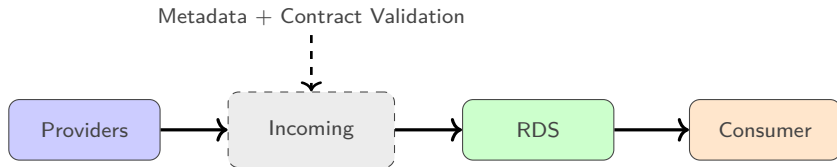
② Cultural Success Factors

- Data should be treated as business assets
- Thinking should be at the enterprise level
- Accept and follow the governance practices

How is Data Architecture Implemented

Data Architecture Implementation Process:

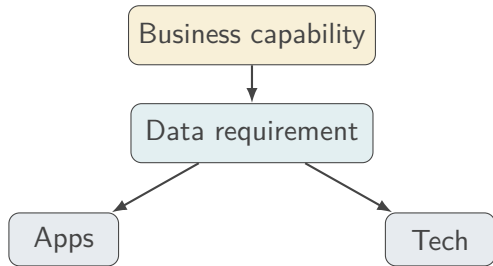
- Data is received from **data providers (source systems)**
- Data enters an **incoming tier (landing zone)**
- **Metadata is used** to validate schema compatibility
- Validated data is stored in **data storage (e.g., RDS)**
- Approved data is made available to **data consumers**
- Ensures **controlled, consistent, and trusted data flow**



Integrating data architecture with enterprise architecture

- Data architecture must be integrated, not isolated
- Business capabilities drive data dependencies
- Application and technology choices must respect data requirements
- Architecture decisions should be visible in enterprise governance forums

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).



Tools and techniques from the chapter

Tools

- Data modeling tools
- Asset management software
- Graphical design applications

Techniques

- Lifecycle projections
- Diagramming clarity

Simple message

The tool does not create architecture quality by itself. Clear diagrams, consistent semantics, and disciplined lifecycle thinking do.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Implementation guidelines and common risks

- DAMA explicitly highlights **readiness assessment** and **risk assessment**
- Implementation depends on organization and culture, not only design quality
- Architecture work needs communication, adoption, and behavior change
- Success requires balancing enterprise standards with local project realities

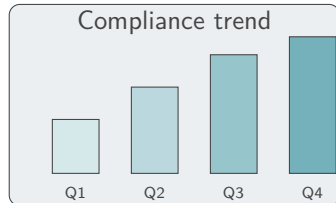
Typical risks

Poor buy-in, unclear ownership, weak standards adoption, over-complex diagrams, and projects bypassing enterprise architecture.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

- Data architecture needs governance to stay relevant and enforced
- Standards compliance is a major governance concern
- Metrics in the chapter include:
 - architecture standards compliance rates
 - trends in implementation
 - business value metrics

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).



Real-world style examples by section

Section		Example
Business drivers		Automotive firm adds connected-car services and now needs operational vehicle data as a managed asset
Enterprise data model		Insurer defines Party, Policy, Claim, Payment, and Product as shared enterprise concepts
Subject areas		University separates Student, Course, Enrollment, Finance, and Identity but keeps them related
Data flows		Hospital maps lab results from source system to EHR, billing, and analytics environments
Governance metrics	&	Retail group tracks compliance of new project schemas with enterprise customer definitions

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Scenario

A company wants speed. Project teams want to copy customer data into every application because it is easier in the short term.

- What short-term benefit are they chasing?
- What long-term data architecture risks appear?
- Where would you allow duplication, and where would you not?

Hint: Use the language of trust, redundancy, lineage, integration cost, and ownership.

Key takeaways

- ➊ Data architecture is the enterprise **blueprint** for managing data assets.
- ➋ It connects **business strategy** to **data requirements** and then to **technology execution**.
- ➌ Its main artifacts are enterprise data models, subject area models, and data flow designs.
- ➍ It works only when integrated with broader enterprise architecture and governance.
- ➎ Good architecture reduces complexity, improves trust in data, and increases business agility.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Questions, reflections, or examples from your own organization?

Possible prompts:

- Where does your organization struggle most: models, flows, ownership, or standards?
- Which enterprise subject areas are hardest to define consistently?
- What architecture artifact would create the fastest business value for you?

Primary source used for the content structure and DAMA-aligned concepts in this deck:

DAMA International. *DAMA-DMBOK: Data Management Body of Knowledge, Second Edition*. Technics Publications, 2017. Chapter 4: Data Architecture.

Slides in this deck specifically reflect the chapter's definition of Data Architecture, its business drivers, architecture domains, enterprise data models, subject area models, data flows, activities, tools, techniques, implementation guidance, governance, and metrics.